Welcome to the Black Swan Research Project!

Team Introduction:

Welcome to the Black Swan Research project here at Shepherd Venture. Over the past seven years, our team has dedicated itself to understanding and managing the unpredictable nature of stock market crashes—events we liken to 'Black Swans.' Inspired by the medical field's approach to virus immunization, we have developed a groundbreaking 'vaccine' for the 'Black Swan' market events. We've identified a unique dataset which is called 'Swan Beta', it follows a Gaussian distribution, enabling us to apply advanced tools like K-Means Clustering, Mean-Variance Optimization (MVO), and other machine learning skills to manage portfolio risks effectively.

Our research has cataloged 14 significant Black Swan events over the last 50 years, analyzing their patterns and impacts. We aim to let the model have three main functions: Anticipate potential downturns, position the portfolio to minimize losses in the black swan market, and optimize returns during such times.

As new interns on our team, you'll be asked about this model's ongoing development and refinement. You will help us test hypotheses, analyze data, and develop new strategies to enhance our 'vaccination' approach.

Intern Roles and Responsibilities:

As part of the Black Swan Research project, you will be entrusted with a range of responsibilities crucial to our ongoing efforts' success.

1) Model Testing and Refinement: Applying theoretical knowledge, you will test the existing models under various market scenarios. This includes stress-testing our 'vaccinated' portfolios against historical and simulated market downturns to evaluate their resilience.
2) Research and Documentation: You will assist in documenting our findings and contributing to the research project. This will involve reviewing literature, synthesizing information, and clearly articulating our research progress and outcomes.
3) Development of New Strategies: You will ask to do original research, we encourage you to propose and develop new strategies for data analysis, data forecasting, and portfolio optimization.
4) Collaboration and Communication: This is a remote position, and we ask you to attend the team meeting every Thursday 3:00 PM PST. Submit your research report which contains your research progress weekly (Every Wednesday by noon PST).

Methodology and Tools:
In this project, we apply a detailed analysis of "Swan Beta" values using K-means clustering and optimizing portfolio allocation through Mean Variance Optimization (MVO). Here's some key metrics and methodologies:

1) K-means Clustering for Swan Beta Values: We use K-means clustering to segment stocks based on their Swan Beta values, which measure the impact of market downturns on individual stocks relative to the market. By clustering these Betas, we can identify groups of stocks that behave similarly during Black Swan events.

2) Forecasting Next Beta Values: After clustering the stocks, we focus on forecasting the next potential Beta values for each stock. Stocks with the lower forecasted Beta values are considered safer and are selected for the next step in our portfolio management process. (Currently, we use Holt Estimators as the best method to forecast future beta values for each stock.)

3) Mean Variance Optimization (MVO): With the identified stocks from the lowest Beta cluster, we apply MVO to allocate these stocks within a portfolio optimally.

4) Tools: Our analytical work is performed using Python, and Excel remains our primary tool for presenting and visualizing the outcomes of our models. We use Excel to demonstrate the effectiveness of our portfolio strategies to stakeholders by showcasing the risk-return profiles, Beta distributions, and other relevant financial metrics.

5) Data for stocks. Currently, we use Yahoo Finance for current and historical stock prices. Historical data is stored in OPTDAT files on Google Drive and GitHub.

Useful Videos:
CAPM: This video discusses what CAPM is, what its calculation entails, and how it can be used, given two very simple examples.
Sharpe Ratio: This video discusses the Sharpe Ratio, how it is calculated, and a scale that interprets what a given ratio means.
MVO: In a quick video, MVO is discussed and how it went against what was previously thought to work for investing, creating an innovation in Portfolio Theory.
K-Means: This video gives an overview of k-Means clustering and its process by giving clear graphical representations.
Also: https://www.youtube.com/watch?v=_aWzGGNrcic&list=LL&index=5&pp=gAQBiAQB8AUB